Deception detection and recovery in model-free NE seeking

Working notes

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The whole thing in one line. All three results use the same primitive: *demodulate at a frequency the deceiver isn't injecting into.* Use it to *compare* and you get detection; use it to *update* and you get recovery. Each result also has a boundary, and knowing where they are is most of the contribution.

1 Setup

Victim v, deceiver d, scalar actions, costs $J_v(x_v, x_d)$ and $J_d(x_v, x_d)$. Both carry sinusoidal probes:

$$x_v = u_v + a\sin(\omega_v t) \;\big[+\; a\sin(\omega_v' t)\big], \tag{1}$$

$$x_d = u_d + a\sin(\omega_d t) + a\,\delta_d(t)\sin(\omega_v t). \tag{2}$$

The attack is the last term of (2): the deceiver injects at the victim's *own* probe frequency. The bracketed second probe enters the action but drives no update — only the detector uses it.

Payoffs go through a washout (high-pass) filter with corner ω_h, then get demodulated:

$$\dot z_i = -\omega_h(z_i - J_i), \quad h_i := J_i - z_i; \qquad \dot u_v = -\tfrac{2k}{a}\,\overline{h_v \sin(\omega_v t)}, \qquad \dot\delta_d = \varepsilon\big(J_d - J_d^{\mathrm{ref}}\big). \tag{3}$$

Setting $\delta_d \equiv 0$ recovers honest NE seeking.

DNE. The deceptive equilibrium (u^*, δ^*) solves

$$\underbrace{\partial_v J_v(u^*) + \delta^* \partial_d J_v(u^*) = 0}_{\text{victim } \textit{thinks} \text{ it's optimal}}, \qquad \partial_d J_d(u^*) = 0, \qquad J_d(u^*) = J_d^{\mathrm{ref}}. \tag{4}$$

Standing assumptions. $J_v, J_d \in C^3$; probe frequencies distinct and non-resonant ($m_1\omega_v + m_2\omega_v' + m_3\omega_d \neq 0$ for $0 < |m| \le 3$), with $k \ll \omega_h \ll \omega_{\min}$; *probe secrecy* — the deceiver knows ω_v and injects there only; *bounded injection* $|\delta_d| \le \bar\delta$. The last one is vacuous while deception succeeds and bites only after recovery (§7).

1.1 Why washout isn't optional

Expand J_v about u: you get $J_v(u)$ at $O(1)$ plus the signal $a\sin(\omega_v t)\partial_v J_v$ at $O(a)$. Multiplying by $-\frac{2k}{a}\sin(\omega_v t)$ creates an asymmetry — the signal carries its own a which cancels the $1/a$, the DC term doesn't.

"It averages to zero" doesn't save you, because u_v *integrates* it, and $\int \sin = -\frac{1}{\omega}\cos$. The result is a ripple of amplitude $2k|J_v|/(a\omega_v)$ riding on u_v. What matters is ripple versus probe:

$$\frac{2k|J_v|}{a\omega_v} < a \iff \omega_v > \frac{2k|J_v|}{a^2}.$$

At our parameters ($k = 0.3$, $a = 0.05$, $J_v \approx 0.59$, $\omega_v = 2\pi \cdot 5$) the ripple is 0.225 against a probe of 0.05 — **4.5× larger**, and the loop destabilises. Fixing it by raising ω needs ∼225 Hz instead of 5 Hz, and the a^{-2} means halving a quadruples the requirement.

Washout tracks and subtracts the slowly-varying $J_v(u)$, leaving an $O(a)$ signal. Ripple drops one order to k/ω, the requirement relaxes from a^{-2} to a^{-1}, and the ripple/probe ratio becomes 0.19 instead of 4.5.

Washout doesn't kill the ripple, it demotes it. What's left is the *relative* ripple

$$\nu := \frac{k}{a\,\omega_{\min}} = 0.191,$$

and $0.19\nu^2 = 0.0068$ is exactly the honest noise floor we measure (0.00683). This ν comes back in §8 as the real design variable.

2 Lemma 1: everything rests on this

$$\text{(a)} \quad C_v = \partial_v J_v + \delta_d\, \partial_d J_v + O(a^2), \qquad \text{(b)} \quad C_v' = \partial_v J_v + O(a^2). \tag{5}$$

Demodulating at ω_v, the injection is at the same frequency and $\overline{\sin^2} = \frac{1}{2}$ keeps it. At ω_v', $\overline{\sin(\omega_v t)\sin(\omega_v' t)} = 0$ and it vanishes. **The only difference between the two channels is whether the injection is there.**

3 Result 1 — blindness

At the DNE, $C_v = O(a^2)$. Proof: substitute the first line of (4) into (5)(a); the bracket is zero by definition.

So the victim staring at its own update signal cannot tell the two regimes apart. Its true gradient $\partial_v J_v(u^*) = -0.2045$ is nowhere near zero, but its *perceived* gradient is. The action residual is equally blind: 1.6×10^{-2} honest vs 1.5×10^{-2} deceptive.

Don't over-read the numerics. $|C_v|$ measures 1.5×10^{-13} honest against 6.1×10^{-5} deceptive — eight orders apart, which looks exploitable. It isn't. Both sit far below the $O(a^2) = 2.5 \times 10^{-3}$ floor they're only guaranteed to respect, so the gap is an artifact of noiseless integration. At $\sigma = 10^{-4}$ they become 4.4×10^{-5} and 6.2×10^{-5} — a ratio of 1.4 — while D_v is unchanged at 0.0068 vs 0.2057.

4 Result 2 — detection

Take a second demodulation at ω_v' and set $D_v := |C_v' - C_v|$. Subtracting (5)(a) from (b):

$$\boxed{D_v = |\delta_d| \left|\partial_d J_v(u)\right| + O(a^2) \quad \text{at every time } t} \tag{6}$$

The true gradient appears in both channels and cancels, leaving exactly the injection term C_v couldn't isolate.

Why the difference, not C_v' alone. They agree at the DNE, since $C_v = O(a^2)$ there. They don't agree during honest transients, where $C_v' = C_v \neq 0$ — so testing $|C_v'|$ alone raises an alarm on *every honest run that hasn't converged yet*. The subtraction is what buys "at every time t" rather than "after settling". It's also sharper: each channel carries a bias around 0.010, but their difference has a floor of 0.0068, because the biases are largely common-mode.

Threshold. With $\eta := \sup_t \left|D_v - |\delta_d||\partial_d J_v|\right|$ the total residual, if $\eta \leq \frac{1}{4}\delta_{\min} c_0$ and $\rho := \frac{1}{2}\delta_{\min} c_0$, then the rule "$D_v > \rho \Rightarrow$ deception" has neither false alarms nor misses — against a deceiver that respects secrecy.

D_v uses only the victim's own payoff samples and its own probes. No model of J_v, no knowledge of δ_d, J_d, or who the deceiver is. **Same model-free setting as the NE seeking itself.**

5 The attack surface

These two go together. One asks what an honest-ish deceiver can get away with; the other asks what a cheating deceiver can do.

5.1 Deceivers that respect secrecy: greed is loud

Since $D_v \geq c_0|\delta| + O(a^2)$ while the gain $\pi(\delta)$ is continuous with $\pi(0) = 0$, any deceiver the detector reads as honest is confined to $\bar{\pi}(\rho) \to 0$ as $\rho \to 0$.

The mechanism is simple: the same δ_d that makes you money is what makes you visible. So the threshold doesn't just classify, it *prices*.

Numerically, sweeping J_d^{ref} traces a frontier where profit and D_v are monotonically related with $r = 0.9998$, and the quietest attack that stays under $\rho = 0.03$ recovers only **4.5% of the honest payoff**.

5.2 Deceivers that don't: exact evasion

If the deceiver injects on both probes, adding $a\delta'(t)\sin(\omega_v' t)$ to (2), then

$$D_v = \left|\delta' - \delta_d\right| \left|\partial_d J_v(u)\right| + O(a^2). \tag{7}$$

Choosing $\delta' \equiv \delta_d$ gives $D_v = O(a^2)$: **the detector reads honest while the attack runs at full strength.** And it's free to first order — the victim updates only at ω_v, so the extra component averages out of $\dot{u}_v$ and the DNE is unchanged to $O(a^2)$.

Think of two scales, one possibly tampered with and one clean, compared against each other. The deceiver tampers with both identically and the difference is zero again.

Writing $\delta' = \mu\delta_d$, (7) predicts a V with vertex at $\mu = 1$ and slope $|\delta_d \partial_d J_v| \approx 0.200$:

μ	0	0.25	0.5	0.75	**1**	1.25	1.5	2
D_v	0.2057	0.1559	0.1060	0.0561	**0.0061**	0.0440	0.0943	0.1952
DNE u_v	0.5424	0.5423	0.5421	0.5419	0.5417	0.5415	0.5412	0.5406

At $\mu = 1$ the detector reads 0.0061, *below* the honest floor of 0.00683, and the DNE has moved by 6.6×10^{-4}. Measured slopes: 0.1996 and 0.1891.

So the guarantee is probe secrecy, not unconditional soundness. It is not "impossible to fool" — it is "you can't fool it *unless you learn* ω_v'". The strength of the result equals the victim's ability to keep ω_v' secret.

Order of presentation matters: lead with the evasion, follow immediately with the pricing result. The other way round it reads as a confession.

6 Secrecy is a timing condition

The deceiver's payoff depends on x_v through $\partial_v J_d \neq 0$ — which is precisely what makes deception profitable. But x_v carries the victim's probes, so *the deceiver's own payoff contains an* ω_v' *component.* It can't both profit and be deaf to the probe. Spectrally scanning its own payoff recovers ω_v' after an acquisition time T_{acq} set by its measurement noise and by a.

So secrecy is only realistic if the victim re-draws ω_v' with period T_{hop} and

$$\underbrace{\frac{1}{\omega_{\min}}}_{\text{probe period}} \ll \underbrace{T}_{\text{detection window}} \ll \underbrace{T_{\text{hop}}}_{\text{re-draw period}} < \underbrace{T_{\text{acq}}}_{\text{time to crack}} . \tag{8}$$

The middle inequality says detection must finish between re-draws; the right one says the deceiver must not finish first. T_{acq} **is not addressed** — it needs an observation model for the deceiver, which is a separate piece of machinery.

Hopping is free, stability-wise. Re-drawing makes the victim a switched system, while the recovery theorem only analyses one switch. You'd expect a dwell-time bound, and it would collide with $T_{\text{hop}} < T_{\text{acq}}$. It doesn't appear. Cycling through five admissible frequencies:

dwell (s)	20	5	1	0.2	**0.05**
recovery error	1.5×10^{-3}	2.0×10^{-3}	3.1×10^{-3}	4.2×10^{-3}	$\mathbf{1.9\times10^{-2}}$

The shortest dwell is a third of the washout time constant $\omega_h^{-1} = 0.159\,\mathrm{s}$ and 1/67 of $k^{-1} = 3.33\,\mathrm{s}$, amounting to **3600 switches in one run**, with no divergence anywhere and everything still inside the $O(a) = 0.05$ band. The victim's slow dynamics average over the switching.

This is a negative result, and a useful one: it removes an objection rather than adding a hypothesis. The binding constraint on T_{hop} is the detection window, not stability.

7 Result 3 — recovery

Detection and recovery are two uses of one primitive. At the trigger time, re-randomize the frequency driving the update from ω_v to a secret ω_v''.

The victim now demodulates at ω_v'' while the deceiver still injects at ω_v, so the cross term carries $\overline{\sin(\omega_v t)\sin(\omega_v'' t)} = 0$. The perceived gradient loses its bias, $\dot{u} = -kG_{\mathrm{hon}}(u) + O(a^2)$, and u converges exponentially to an $O(a)$-neighbourhood of the true NE.

Note δ_d has *decoupled* from the u-dynamics — but decoupling alone doesn't make it irrelevant. The neglected Taylor remainders carry $a^2\delta_d^2\partial_{dd}^2 J_v$ and $a^3\delta_d^3\partial^3 J_v$, so an unbounded δ_d would destroy both the perturbation bound and the containment of x_d in U. That's what the bounded-injection assumption is for.

7.1 What happens to the deceiver afterwards

If it runs its integrator without saturation, then since $J_d(u(t)) \to J_d(u^\star)$ the integrand tends to a constant $\gamma := J_d(u^\star) - J_d^{\mathrm{ref}} \neq 0$, so

$$|\delta_d(t)| \;\geq\; |\delta_d(t_1)| + \theta\,\varepsilon|\gamma|\,(t - t_1). \tag{9}$$

Linear divergence at rate $\varepsilon\gamma = 4.33 \times 10^{-3}$; measured 4.68×10^{-3} (8.1%). At actuation limits $a|\delta_d| \leq 0.5, 1, 2$ that's roughly 2.4×10^3, 4.5×10^3, 8.8×10^3 time units to visibility.

deceiver behaviour	final u_v	error to $u^\star$	final δ_d
unsaturated	0.7714	2.15×10^{-3}	3.5291 (diverging)
saturated at $\bar{\delta} = 0.5$	0.7712	2.01×10^{-3}	0.5000
abandons ($\varepsilon = 0$)	0.7712	2.00×10^{-3}	0

An unbounded injection at a *known* frequency is visible to any spectral monitor; a saturated one satisfies the assumption and the theorem applies verbatim, with the injection persisting uselessly at a frequency nobody demodulates; abandoning leaves an honest game. All three end the deception, and the victim's final error is the same to within 2×10^{-4}. The asymmetry the deceiver exploited — one influence channel at a frequency the victim was obliged to keep using — is gone the moment the victim can move.

On projecting onto a compact set. The natural fix is to project δ, as in arXiv:2407.05168. Doing that loses the corollary above: treating unboundedness as a bug to be patched hides the conclusion that recovery *terminates* the deception. But projection *is* a constructive implementation of the bounded-injection assumption — the difference is only whether it lives in the

algorithm or in the hypothesis. Both presentations work and could be given side by side.

7.2 The switch target is not arbitrary

The proof only needs $\overline{\sin\omega_v t \sin\omega_v'' t} = 0$, which any distinct frequency gives. The higher-order terms are what bite:

$\omega_v''/2\pi$ (Hz)	smallest $\lvert m\rvert \le 3$ combination	resonant?	error to $u^\star$
16.4	1.80	no	0.0014
$15.0 = 3\omega_v$	0.40	only at $\lvert m\rvert = 4$	0.0016
13.9	0.70	no	0.0021
$10.0 = 2\omega_v$	0, at $m = (2, -1, 0)$	**yes**	0.0059
$12.3 = \omega_v + \omega_d$	0, at $m = (1, -1, 1)$	**yes**	**0.0267**

The bound $|m| \le 3$ is tight both ways: relaxing it admits targets costing 4× and 19×, tightening it to $|m| \le 4$ would exclude a target that costs nothing.

8 The design variable is k/ω, not a

Reading (6) naively suggests taking a small. That's backwards. The residual splits into a curvature bias $O(a^2)$ and a ripple bias $O(\nu^2)$ with $\nu = k/(a\omega_{\min})$, which *grows* as a shrinks:

$$\eta \lesssim C_1 a^2 + C_2\Big(\frac{k}{a\,\omega_{\min}}\Big)^2, \qquad a^\star = \Big(\frac{C_2}{C_1}\Big)^{1/4}\sqrt{\frac{k}{\omega_{\min}}}. \tag{10}$$

So the design condition is a condition on $k/\omega_{\min}$, not on a alone.

a	ν	η	η/ν^2
0.025	0.3820	**0.02739**	0.188
0.050	0.1910	0.00683	0.187
0.100	0.0955	0.00171	0.187
0.150	0.0637	0.00076	0.187

Fit: $\eta \approx 0.191(k/a\omega_{\min})^2$ with 1.9% spread across a sweep of a, k and an overall frequency scale; $C_1 a^2$ stays below resolution throughout. **Halving a from 0.05 to 0.025 pushes η from 0.00683 to 0.02739 — nearly onto the threshold $\rho = 0.03$.** Over this range a should be increased, not decreased.

Finite window. A sliding window $\frac{1}{T}\int_{t-T}^{t}$ adds ripple $O(1/(\omega_{\min}T))$ but needs $T \ll 1/k$ to keep u frozen. The two pull opposite ways and they are *not* symmetric: the lower bound is hard (ripple above the floor produces false alarms outright), the upper bound is soft (it degrades transient fidelity, gradually). Here $k/\omega_{\min} = 9.5 \times 10^{-3}$ and the nominal interval $(0.03, 3.3)$ s is two decades wide — yet *every* window in it produces false alarms, and the smallest usable one is $T = 10\,\text{s} = 3k^{-1}$. Transient error is smallest at $T = 10$ and has only doubled at $T = 40$.

9 Numerics

Everything integrates the true time-varying system by RK4, never the averaged one — the averaged system satisfies the identities by construction and would prove nothing. Base game $J_v = \frac{1}{2}(x_v - 1)^2 + cx_vx_d + ex_d$, $J_d = \frac{1}{2}(x_d - 1)^2 + cx_vx_d$, $c = 0.3$, $e = 0.5$; $a = 0.05$, $k = 0.30$, $\varepsilon = 0.08$, probes $2\pi\{5, 7.3, 11.7\}$, $\omega_h = 2\pi$, $J_d^{\text{ref}} = 0.15$, $T = 10$, $\rho = 0.03$.

Claim	Result	Verdict
analytic DNE self-consistent	$\delta^* = 0.30834$, $D_v^{\text{pred}} = 0.20454$	ground truth
DNE where theory says	$\lvert\Delta u_v\rvert = 2.09 \times 10^{-3}$	pass, $O(a^2)$
blindness of C_v	1.48×10^{-13} vs 6.13×10^{-5}	blind as claimed
Lemma 1(a)/(b) residuals	0.01139 / 0.00981	pass
detection identity	0.20571 vs 0.20454 (0.57%)	pass
threshold separation	$0.00683 < 0.03 < 0.20519$	holds, 30×
cost of second probe	DNE shifts 5.38×10^{-6}	negligible
mirroring evasion	$D_v : 0.20571 \to 0.00609$	**detector defeated**
windup rate	0.00468 vs $\varepsilon\gamma = 0.00433$	pass, 8.1%
recovery after hop	$u_v = 0.7714$ vs $u^\star = 0.7692$	pass
repeated hopping	stable at dwell 0.05 s, 3600 switches	no dwell limit
residual scaling	$\eta \approx 0.19(k/a\omega)^2$	corrects the corollary
measurement noise	FA rate 25.7% at $\sigma = 0.05$	**open**
profit vs detectability	$r = 0.9998$	monotone
worst hop target	12.3 Hz, error 0.0267	admissible set matters
non-quadratic game	0.18712 vs predicted 0.18922 (1.11%)	pass

Two things worth flagging. The identity holds *along the trajectory*, not just in the limit: pointwise error averages 0.0035 over the transient and 0.0056 at steady state. And the cubic variant ($J_v + \frac{q}{3}x_v^3$, $q = 0.4$) is what actually loads the third-order remainder — a quadratic game has $\partial^3 J \equiv 0$, so it was never under test before.

10 Open

1. T_{acq}**.** The right end of (8). Needs an observation model for the deceiver — noise, scan strategy, estimator. Most likely to become its own paper.
2. **Randomized probe bank.** Give the victim m monitoring probes and draw one at random. Direct hardening against (7); not yet quantified.
3. **Noise-dependent threshold.** 25.7% false alarms at $\sigma = 0.05$ with a fixed threshold. Needs calibration or an adaptive test (CUSUM / GLRT). Reviewers will hit this.
4. **Projection variant.** Whether to present it alongside the windup corollary.
5. **Vector / N-player.** Everything carries over with Jacobian blocks and norms, but note that evasion becomes $(\nabla_d J_v)(\delta' - \delta) = 0$ — for $n_v < n_d$ this admits solutions with $\delta' \neq \delta$, i.e. the deceiver only has to match the injection modulo $\ker(\nabla_d J_v)$. Currently a sketch.